# What to build first — v1 / Phase 3

*Q: Which piece do we build next, what does it unlock, and how do we know it is done?*

REQ-01 REQ-04 REQ-05 REQ-07

| INPUT · WHERE THE DATA COMES FROM | BUILD THIS, IN THIS ORDER | DONE WHEN · observable, not a feeling |
|---|---|---|
| nothing external yet —<br>no GitHub, no git, no disk | **1 API SKELETON** ~1 h<br>POST /analyze + GET /analysis/{analysis_id}<br>return a hardcoded but schema-valid AnalysisResult<br>**follow: M4** | Their tester can reach you.<br>You start reading real failures<br>instead of guessing which to fix. |
| two regexes<br>`^[A-Za-z0-9][A-Za-z0-9-]{0,38}/[\w.-]{1,100}$`<br>`^[0-9a-fA-F]{7,40}$` | **2 VALIDATE THE REQUEST** ~1 h<br>repository matches owner/name · commit_id is 7–40 hex<br>anything else -> 400 INVALID_REQUEST, nothing saved<br>**follow: M2 · VALIDATING** | POST {"repository":"not-a-repo"}<br>returns 400 INVALID_REQUEST<br>and no analysis_id is issued. |
| `GET /repos/{owner}/{repo}`<br>`/commits/{sha}`<br>`-> .sha (full 40)`<br>`-> .parents[0].sha` | **3 RESOLVE THE COMMITS** ~3 h<br>head := commit_id expanded to the full 40-hex SHA<br>base := base_commit, else the first parent of head<br>**follow: M2 · RESOLVING** | Send 2a8a38b, get back the full<br>40 chars and base d8eaaba8…<br>A root commit gives base = null. |
| `GET /repos/{owner}/{repo}`<br>`/compare/{base}...{head}`<br>`-> .files[] filename`<br>`status`<br>`previous_filename` | **4 GET THE DIFF** ~4 h<br>changes[] with path · old_path · change_type<br>GitHub's status vocabulary maps 1:1 onto the enum<br>**follow: M4 · Change** | Rename yields ONE change with<br>old_path set, not a delete + add.<br>T-01 T-02 T-03 T-06 T-07 pass. |
| `docs/ *.md *.rst`<br>`-> DOCUMENTATION`<br>`src/ tests/`<br>`-> IMPLEMENTATION`<br>`no rule -> OTHER` | **5 CLASSIFY EACH FILE** ~2 h<br>category per changed file · first matching path rule wins<br>hardcode the rules now; mapping.json swaps the innards later<br>**follow: M3 · CLASSIFY box only** | docs/installation.rst comes back<br>DOCUMENTATION, not OTHER.<br>T-04 passes. |
| local disk or sqlite.<br>storage "local" is fine for v1 —<br>note that Render wipes it<br>on every redeploy. | **6 SAVE AND RETURN** ~3 h<br>write runs/{analysis_id}.json · GET returns it byte-identical<br>binaries and lockfiles -> skipped[], status stays SUCCESS<br>**follow: M2 · SAVING + M4** | Three identical runs differ only<br>in analysis_id and created_at.<br>AC-06, AC-10, T-05 pass. ~17/23. |

## DO NOT BUILD YET

*Each of these is a later phase. Starting one now costs you Phase 3.*

- X **The M3 finding rule**
  MISSING_ / OUTDATED_DOCUMENTATION. Phase 4.
  *issues[] staying empty is CORRECT for v1, not a bug.*
- X **mapping.json loader + config_version hash**
  End of Phase 3 into Phase 4. This is component C4.
  *Until it exists, return config_version = null — never a constant.*
- X **M5 review workflow, PR comments**
  Phase 5. This is component C7.
- X **heading_path, :: selectors, tree-sitter**
  Schema Tier 2. Not in the common scope at all.

## WATCH OUT · these four will cost you a day each

- ! **GitHub unauthenticated is 60 requests per HOUR.**
  One analysis costs 2–4 calls. You will exhaust it mid-debug and think your code broke.
  Use a fine-grained PAT with public-repo read -> 5,000/hour.
- ! **Build to the v1.1 field names, not M4-as-originally-written.**
  path, old_path, category — not file_path / change_category.
  Their tester encodes these. The old names fail T-03 and it will look like your bug.
- ! **Render free tier sleeps.**
  Warm is ~3 s; cold can exceed 50 s.
  Set generous timeouts in BOTH testers or you will log failures that are just sleeping dynos.
- ! **The compare API caps at 300 files.**
  Fine for pod fixtures; a large real repo will silently truncate.
  Worth a note in your README now, a paginated fallback later.

Everything in this model is components C1, C2, C3, C5 and C6. C4 (mapping loader) and C7 (detection + review) are deliberately outside this slice — which is exactly the coverage the other team's service reports today. Finishing these six steps puts you level with them, on a contract you both share.